

# Parameterized Pipelined-CORDIC Softmax IP

## Functional Verification Report

### 1. Executive Summary

This report summarizes the IP-level functional verification of a parameterized pipelined-CORDIC Softmax accelerator. The verification environment was developed using SystemVerilog and UVM methodology with constrained-random stimulus, mathematical reference modeling, transaction-level scoreboard checking, and coverage-driven verification closure.

Verification Status: PASS - Functional verification closure achieved.

Regression scope:

- Total tests: 24
- Parameter configurations: NUM\_ELEMENTS = 4, 8, 13, 16
- Verification methodology: SystemVerilog + UVM
- Simulator: Synopsys VCS V-2023.12-SP2

### 2. Verification Environment

The verification environment consists of an active input agent, input/output monitors, a double-precision Softmax reference model, FIFO-based scoreboard, functional coverage collector, and regression infrastructure.

The scoreboard performs per-element fixed-point comparison between DUT output and expected results generated by the mathematical model.

### 3. Regression and Parameter Coverage

The regression campaign covers four parameter configurations: NUM\_ELEMENTS=4, 8, 13 and 16.

The 13-lane configuration specifically validates non-power-of-two parameter support, including tree padding behavior and parameter-dependent latency.

### 4. Coverage Closure Summary

Coverage closure was performed using Synopsys URG with merged coverage database collected from all regression configurations.

The merged database represents the complete parameter space rather than a single DUT configuration.

| Coverage Metric           | Merged Result |
|---------------------------|---------------|
| URG Score                 | 88.84%        |
| Line Coverage             | 99.30%        |
| Condition Coverage        | 97.96%        |
| Branch Coverage           | 97.77%        |
| Toggle Coverage           | 57.92%        |
| Functional Coverage Group | 91.25%        |

## Appendix A: Merged URG Coverage Dashboard

### Total Coverage Summary

| SCORE | LINE  | COND  | TOGGLE | FSM | BRANCH | GROUP |
|-------|-------|-------|--------|-----|--------|-------|
| 88.84 | 99.30 | 97.96 | 57.92  |     | 97.77  | 91.25 |

### Hierarchical coverage data for top-level instances

| SCORE | LINE  | COND  | TOGGLE | FSM | BRANCH | NAME   |
|-------|-------|-------|--------|-----|--------|--------|
| 88.24 | 99.30 | 97.96 | 57.92  |     | 97.77  | tb_top |

### Total Module Definition Coverage Summary

| SCORE | LINE  | COND  | TOGGLE | FSM | BRANCH |
|-------|-------|-------|--------|-----|--------|
| 76.76 | 94.16 | 80.00 | 46.01  |     | 86.89  |

### Total Groups Coverage Summary

| SCORE | INST SCORE | WEIGHT |
|-------|------------|--------|
| 91.25 | 91.25      | 1      |

Coverage holes are reviewed according to design reachability and verification intent. Remaining uncovered items are classified as unreachable scenarios, structural constants, or non-functional toggle limitations and are documented through coverage waiver analysis.